MIL-M-50277A(PA)
30 September 1974
SUPERSEDING
MIL-M-50277(MU)
3 April 1969

MILITARY SPECIFICATION

MODIFICATION TOOL KIT, ELECTRICAL CONNECTOR

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of electrical connector modification tool kit which is used with the Honest John warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1168	- Lot Numbering of Ammunition.
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components.

DRAWINGS

ORDNANCE CORPS

8880694	- Modification Tool H4225.
8881264	- Universal Joint.
8881272	- Cutter.
8881307	- Retainer, Apron.
8881308	- Spacer, Apron.
8881309	- Apron.
8881311	- Gage Assembly.
8881318	- Cutter Safety Plug Holder,
8881335	- Modification Tool Kit, Electrical Connector.
ET-8881335	- Equipment Tabulation.
8881349	- Brush, Chip.
8881350	- Protective Cap Assembly.
8881366	- Finishing Tool Assembly.
8881369	- Cutter Finishing.

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(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The modification tool kit shall comply with all requirements specified on drawing (dwg) 8881335, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. The contractor shall assign serial numbers as required by drawings. The contractor may select any number which complies with the drawing requirement. These numbers shall not be duplicated.

3.5 Functioning. The modification tool H4225 with a maximum torque of 30 foot pounds applied to the universal joint shall cut connector receptacle keys to the requirement specified on dwg 8813512 when used in conjunction with finishing tool assembly dwg 8881366 and tested as specified in 4.4.1.

3.6 Workmanship.

3.6.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.6.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.6.3 Burrs. Parts shall be free from burrs.

3.6.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.6.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise speci-

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fied in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 1 complete assembly (see dwg 8813512) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table I. Tests shall be conducted in order specified in Table I.

Table I

First article inspection and tests

Inspection or Test	Dest	Non-Dest
1. Classification of characteristics (see 4.3.2.1)		X
2. Functioning (see 3.5)		X

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

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4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table II and Table III), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

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Table III
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Cutter (see dwg 8881272).

Categories	Characterisitcs	Method of Inspection
Critical:	None defined.	
Major:		
	101. Mislocation of mounting holes	Gage
	102. Incorrect thickness	Gage
	103. Evidence of poor workmanship (see 3.6)	Visual
	104. Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:	None defined.	

4.3.2.1.2 Retainer, apron (see dwg 8881307).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Incorrect dimension of hexagon opening	Gage
	102. Incorrect diameter	Gage
	103. Compound on unauthorized surface	Visual
	104. Evidence of poor workmanship (see 3.6)	Visual
	105. Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:	None defined.	

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4.3.2.1.3 Spacer, apron (see dwg 8881308).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Incorrect diameter of holes (2 places)	Gage
102.	True position of small diameter holes incorrect	Gage
103.	Scratch in protective finish with base metal exposed	Visual
104.	Evidence of poor workmanship (see 3.6)	Visual
105.	Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:	None defined.	

4.3.2.1.4 Apron (see dwg 8881309).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Incorrect length	Gage
102.	Incorrect height	Gage
103.	Holes not cut in plastic	Visual
104.	Improper bond	Visual/Manual
105.	Evidence of poor workmanship (see 3.6)	Visual
106.	Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:	None defined.	

4.3.2.1.5 Gage assembly (see dwg 8881311).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Incorrect "GO" gaging depth	Gage
102.	Incorrect "NO GO" gaging depth	Gage
103.	Loose part	Visual/Manual
104.	Missing part	Visual
105.	Damaged part	Visual

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106. Evidence of poor workmanship (see 3.6).....Visual
107. Missing, incorrect and illegible marking.....Visual
108. Objective evidence relating to material missing or improper (see 3.1).....Visual

Minor: None defined.

4.3.2.1.6 Cutter, safety plug holder (see dwg 8881318).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Length	Gage
	102. Missing, improper or illegible marking	Visual
	103. Evidence of poor workmanship (see 3.6)	Visual
	104. Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:	None defined.	

4.3.2.1.7 Brush chip (see dwg 8881349).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Length of brush	Gage
	102. Marking missing or illegible	Visual
	103. Evidence of poor workmanship (see 3.6)	Visual
	104. Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:	None defined.	

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4.3.2.1.8 Protective cap assembly (see dwg 8881350).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Incorrect thickness	Gage
102.	Mislocation of pins	Gage
103.	Evidence of poor workmanship (see 3.6)	Visual
104.	Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:	None defined.	

4.3.2.1.9 Finishing tool assembly (see dwg 8881366).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Parts loose or missing	Visual
102.	Marking incorrect, missing or illegible	Visual
103.	Evidence of poor workmanship (see 3.6)	Visual
104.	Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:	None defined.	

4.3.2.1.10 Cutter, finishing (see dwg 8881369).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Mislocation of mounting holes	Gage
102.	Incorrect thickness	Gage
103.	Evidence of poor workmanship (see 3.6)	Visual
104.	Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:	None defined.	

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4.3.2.1.11 Modification tool H4225 (see dwg 8880694).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Tool fails to fit functional gage	Gage
102.	Evidence of poor workmanship (see 3.6)	Visual
103.	Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:	None defined.	

4.3.2.1.12 Modification tool kit, electrical connector (see dwg 8881335).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Any component missing	Visual
102.	Any component damaged	Visual
103.	Missing, improper and illegible marking	Visual
104.	Evidence of poor workmanship (see 3.6)	Visual
105.	Objective evidence relating to material missing or improper (see 3.1)	Visual
106.	Serialization improper or missing (see 3.4)	Visual
Minor:	None defined.	

4.3.2.1.13 Preparation for delivery (see 5).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Preservation, packing, and packaging not in accordance with MIL-STD-1169(MU)	Visual

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102. Marking not in accordance with MIL-STD-1169.....Visual

Minor:

201. Evidence of poor workmanship (see 3.6).....Visual

4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

Test	Test Classification
a. Functioning	Major

None of these tests are considered destructive and samples so tested may be returned to the lot.

4.3.3.1 Functioning. Each modification tool kit shall be tested in accordance with 4.4.1. If any item fails to comply with 3.5, the item shall be rejected and removed from the lot.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Critical and major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL). ELs for assemblies are listed on the Equipment Tabulation, ET-8881335. When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

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4.4.1 Functioning. The connector receptacle simulating the outside dimensions and material of dwg 8813512 shall be mounted to a fixture which simulates the mounting characteristics of the device. Install the protective cap assembly (dwg 8881350) over the connector. Be sure that the protective cap is flush against the connector receptacle and the black engraved marking is at the 12 o'clock position. Then loosen the three press plate bolts until the threaded end of the bolts are flush with the bottom of the flange on the housing. Rotate the universal joint until the black line within the red background on the cutter body is aligned with the hole in the press plate. Install the modification tool H4225 on the connector receptacle in such a way that the black line is at the 12 o'clock position. Rotate the housing bolts clockwise to thread the bolts into the two tapped holes provided on the test fixture. Tighten bolts securely. Then tighten the three press plate bolts securely. Install torque wrench into the universal joint (dwg 8881264). The torque wrench shall be set to 30 foot pounds and rotated clockwise until the black line, on the cutter body is aligned with the hole in the press plate. Remove the torque wrench. Unscrew the two housing bolts and remove the modification tool from the receptacle connector and remove the loose chips or burrs. Install the finishing tool (dwg 8881366) so that the black line on the finishing tool and the black line on the protective cap are in alignment. Grip the knurled edge of the finishing tool and rotate the tool until the arrow indicator is in the 12 o'clock position. Apply pressure in the direction of the arrow and rotate the tool back and forth over the key until the tool turns freely. Repeat this operation in the 5 o'clock and 7 o'clock positions. Remove the finishing tool and remove the burrs and chips from the connector receptacle. Install the gage assembly (dwg 8881311) on the connector receptacle in such a way that the black line is at the 12 o'clock position. Rotate the gage assembly clockwise and counterclockwise for compliance with the go and no-go requirements on the gage assembly.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The modificationtool kit shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(PA).

5.1.2 Level C. The modification tool kit shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(PA).

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5.2 Packing.

5.2.1 Levels A and C. The modification tool kit shall be packed in accordance with the applicable instructions contained in MIL-STD-1169(MU).

5.3 Marking. Marking shall be in accordance with MIL-STD-1169(MU).

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.
b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-T.

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Preparing activity:
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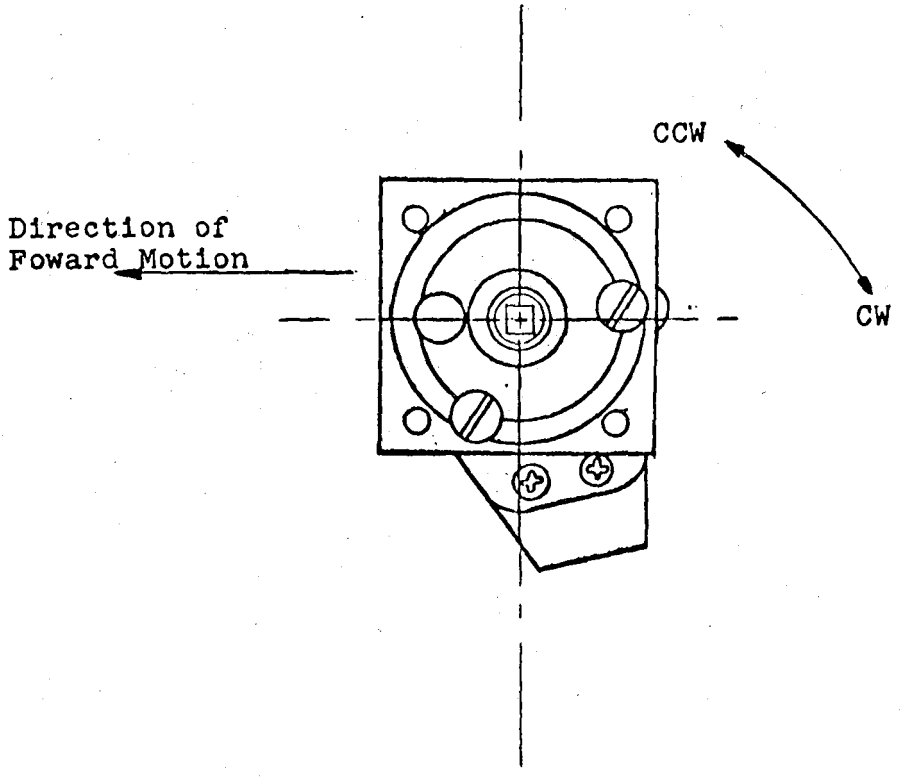


FIGURE 1

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
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DOCUMENT IDENTIFIER AND TITLE

MODIFICATION TOOL KIT, ELECTRICAL CONNECTOR

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 **1426**

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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